

Proof of a Hoof (Where Pixel and Pasture Meet)

Many individuals within the blockchain community are actively exploring ways to effectively connect blockchain with real-world objects. This white paper presents a novel solution that bridges the abstract digital world with the tangible and concrete realm, aiming to solve some of the biggest global environmental problems.

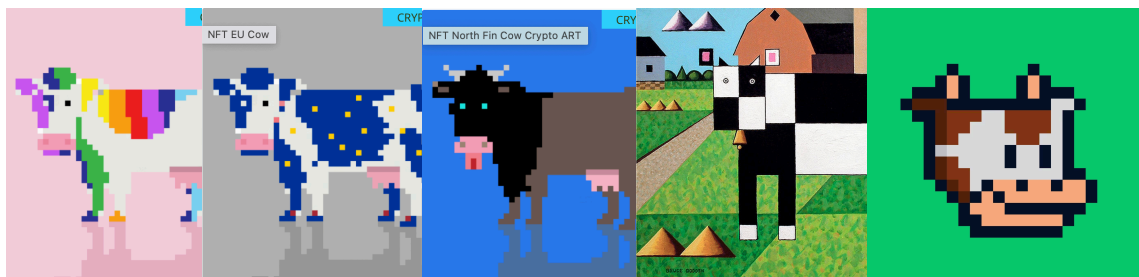
Recognizing the challenge of fully enforcing trust in a decentralized manner, our solution emphasizes shifting the economy toward a more trustworthy environment. This is achieved through a sophisticated system of reputation and charitable staking, empowering local users with global stakeholder support.

Our selection of unique real-world objects, which are initially being connected to digital sphere in form of crypto's NFTs (non-fungible tokens), is driven by a deep consideration of cultural significance. As Yat Siu, an advocate for genuine digital property rights, emphasizes in his article "The Critical Role of Culture and Ownership in the Metaverse":

“Cultural aspects often surpass utility in driving consumption. Consider the realm of fashion, where choices are not solely dictated by practicality but rather serve as expressions of identity and aspirations.”

Amidst the dynamic technological landscape, cultural elements risk being overshadowed by rapidly changing trends. Notably, NFT designs often lack both utility and aesthetic appeal. To anchor our initiative in practicality, commonality, and timeless cultural context, we aim to merge the fast-paced utility of the digital world with the mature day-to-day experiences of the physical world. Obviously the best culture is one which was cultivated for sufficient time.

In recognizing the undeniable interconnectedness of animals with humanity, we strategically chose them as objects—subjects represented by NFTs. To keep the project initially simple and after extensive research, the cow has emerged as the unequivocal symbol of sustainability and ethical responsibility, mainly because of the cow's influence and impact on human society. This integration aims to reshape the narrative around digital assets, positioning NFTs as catalysts for positive social and environmental change. The cow is also chosen for its cultural importance, so much that for example in Hindu religion it is considered sacred. And it is part of day-to-day life of people in almost every country. So cow can become not only a powerful emblem fostering a harmonious coexistence between technology, culture, and the environment but also its protection has basis and is studied deeply by economists and cultural experts.



“According to Hindu mystic mythology taking care of cows provide owner with many material and spiritual gifts manifesting in his/her reality.”

Global problems

Another important problem which this project addresses is the psychological issues people nowadays suffer from, with its high correlation to mindless screen-watching of electronic devices. The profound words of Dr. Sahadeva resonate with our mission: "Because of no contact with the animal world, children are losing contact with nature as they move to a more virtual world. Children today have no clue about nature. Most children in developed countries do not see a live cow or a live horse or an apple tree. How can they protect nature if they do not know it?" (Dasa, Dr. Sahadeva. *Capitalism Communism And Cowism – A New Economics For The 21st Century*. Soul Science University Press. Kindle Edition).

By integrating blockchain into our initiative, we aim to address this disconnection between humanity and nature. The NFTs, represented by a cow, serve not only as a cultural bridge but also as a means to rekindle a connection with the natural world. In our ideal scenario, as blockchain quietly supports the economical independence of self-sustainable communities, people share their experience and knowledge across well protected and authorised chain fostering a generation that understands and values the importance of protecting nature, paving the way for a more harmonious coexistence between technology, culture, and the environment in the 21st century.

Interestingly, cows are often highlighted as major contributors to carbon emissions. This is a complex and technical issue, which will be discussed in more detail later in the paper. However, to illustrate just how *literally bullish* our idea is, let us consider the following question: *Is there an alternative use for cattle beyond beef production?*

One area worth exploring is the looming issue of declining oil resources. Historically, before gasoline became the dominant source of traction power, animals—particularly bulls—were commonly used for labor. Interestingly, studies suggest that bulls actually benefit from moderate physical work, as inactivity can lead to muscle stagnation.

Today, while using bulls in agriculture may seem outdated, their physical power could find novel applications. For example, animal-powered mills could be repurposed for tasks like proof-of-work crypto mining. This might sound unconventional, but there are already projects experimenting with the use of cow dung in biogas-powered crypto mining systems. See Video: [Cows Dung as Proof of Work](#)



Finally, we must recognize the growing global awareness around animal welfare—particularly concerns about slaughter practices and the rise of vegan and ethical-living movements. While we are not engaging in that ethical debate here, it is important to acknowledge the potential for strong support from these large, active communities.

To drive engagement, we will *gamify* this component of the system. Imagine a user “rescuing” a cow from a slaughterhouse—purchasing it with the intention of keeping it as a pet—then documenting the event on the blockchain and using oracle data to geotag the animal’s location. In return, the rescuer earns reputation points, community recognition, and even donations to support ongoing care.

We also envision a playful, family-friendly app—think Pokémon GO in the countryside—where participants of all ages locate, tag, and “scratch” (interact with) real cows, earning points and rewards along the way. This approach not only promotes animal welfare and education but also harnesses blockchain’s transparency to create a truly rewarding, socially meaningful experience.

“Aside from political mobilization for cow protection in India, there is, like nowhere else in the world, a widespread practice of cow care and protection, or sheltering, in a wide variety of settings, including but not limited to institutional goshalas and similar establishments. Not only Hindus, but also Sikhs and Jains are involved in these projects, as are various non-religious animal activist organizations.” (Nagy 2019, pp. 254–257).



Challenge of decentralisation

From both philosophical and practical standpoints, the notion of a *truly* decentralized system is inherently paradoxical. Consider the example of democracy: while it is often perceived as a decentralized political structure, in practice, it functions by empowering individuals to collectively select a central authority. Thus, decentralization exists only in the process, not in the outcome.

A similar pattern can be observed in the natural world. Fundamental forces such as gravity illustrate that all systems tend to organize around focal points. This principle applies equally to digital systems. While it is theoretically possible to envision a network in a state of "voidistic" equilibrium—lacking any dominant nodes or authorities—real-world implementations inevitably evolve toward dynamic asymmetry. Fluctuations in activity, influence, or traffic naturally give rise to centers of gravity within the system.

In this context, blockchain technology does not eliminate centralization but rather reframes it. It provides a decentralized mechanism for the *selection and verification* of central authorities, thereby introducing transparency and robustness into inherently hierarchical structures.

Mechanics

For technical side there is another document, with all the programming and economical issues and solutions presented. Following part represents mainly general social and gamifying dynamics.

Staking and Reputation

So now some practical scenario. When a user mints an NFT representing a real cow, they provide supporting evidence, such as videos, pictures, real-world owner concession and purchase details. This evidence is available to the community before staking/rating occurs. In this way social media serves as oracle itself. NFT is then open for authorised stakers to rate/vote on. The reputation of these stakers determines the power of their ratings, which impacts the overall credibility of the NFT. If reputation of given object/NFT changes, it correspondingly affects all voters staked in given project.

To incentivise whole process, owners or cows, nfts or stakers can convert their hold and reputation into cash, but this results in a loss of reputation. If an NFT is found to be fraudulent or if the cow is not properly cared for, the original staker loses reputation. Conversely, if the cow is well-maintained and the NFT proves to be genuine, the staker's reputation can be positively affected over time. A very significant factor in reputation recovery is how independent the project becomes outside of the stakers' direct involvement. This is crucial for maintaining decentralization. Users are not bound to a particular project; they can initially support it, but as the project becomes more self-sustaining, the staker's control diminishes, although their reputation benefits. The goal of the project is to promote natural decentralization and self-sustainability, with voters incentivized to support projects based on their merit and long-term viability rather than personal gain. The reputation system encourages contributions based on altruism, with financial gains being a secondary benefit.

Similarly, users can convert reputation into cash based on their share of the overall pool. However, this weakens the entire system, which is why withdrawing comes at the cost of losing some of reputation.

Reputation extends beyond individual users to the NFTs themselves and can spread dynamically. High-reputation NFTs and successful projects positively influence the reputation of associated users. Community interactions, including voting on videos, images, and user contributions to discussions, drive this dynamic. Successful collaborations and well-managed projects see their reputation rise, while unsuccessful ones will deteriorate, preventing unwarranted price inflation within the pool.

While the system ensures that reputation is tracked on-chain for transparency and accountability, practical management tasks such as evidence verification and ongoing monitoring are handled off-chain. This approach leverages a self-regulating mechanism where responsible users with high reputation most influentially contribute to the system, maintaining its integrity and functionality. We don't expect many scams because nature of this project is in itself attractive mostly for people who are charitable. The decentralized nature of the blockchain ensures that reputation scores are secure and immutable, while off-chain systems manage the practical aspects of data and project evaluation.

NFT values can be significantly leveraged, with their price increasing based on perceived value and quality. However, continuous proof of proper care is required to maintain the well-being of the cows. Reputation limits prevent price inflation without adequate evidence. If a cow dies, the reputation tied to the NFT can be transferred, but the NFT itself is lost. Users with high reputation can leverage their influence to facilitate the minting of additional NFTs representing other local animals or agricultural fields. This system supports a self-sustaining community, with the overall pool covering payments for herders, farmers, and other contributors."

Is this idea realistically applicable?

While we are providing some basics ideas someone may question if those are just promises or if it's really practically achievable. In answer we are not coming with something which is not already there. Those system are already there they just did not get enough traction , or might not have been used properly. For example following is excerpted from whitepaper of Steemit Blockchain :

"Subjective Proof of Work presents an alternative approach to distributing a currency that improves upon fully objective Proof of Work systems such as mining. The applications of a currency implementing Subjective Proof of Work are far broader than those of any objective system because they can be applied to build a community around any concept with a sufficiently defined purpose.

When individuals join a community, they buy into a particular set of beliefs and can vote to reinforce the community's values or mission. In effect, the criteria by which work is evaluated are entirely subjective, and the definitions of value live outside the source code itself. One community may wish to reward artists, another poets, and another comedians. Others may choose to support charitable causes or promote political agendas. The value of each currency depends on the demand for influence within its respective community and the perceived growth potential of that community.

Unlike prior systems, Subjective Proof of Work enables communities to collectively fund the development of what they find valuable—effectively monetizing time and effort that was previously considered non-monetizable.

Working or Staking

There are two ways people can get involved with a crypto-currency community: they can buy in, or they can work in. In both cases users are adding value to the currency, however, the vast majority of people have more free time than they do spare cash. Imagine the goal of bootstrapping a currency in a poor community with no actual cash but plenty of time. If people can earn money by working for one another then they will bootstrap value through mutual exchange facilitated by a fair accounting/currency system. Distributing a currency to as many people as possible in a manner that is generally perceived as fair is a challenging task. The tasks that can be entirely evaluated by an objective computer algorithm are limited in nature and generally speaking have limited positive external benefits. In the case of Bitcoin-style mining, it can result in the production of specialized hardware and cause people to invest time developing more efficient algorithms. It may even help find prime numbers, but none of these things provide meaningful value to society or the currency holding community at large. More importantly, economies of scale and market forces will end up excluding everyone but experts from participating in this kind of distribution. Ultimately, computation-based mining is just another way of buying in because it requires money to pay the electric bill or the development of hardware necessary to do the work. In order to give everyone an equal opportunity to get involved and earn the currency people must be given an opportunity to work. The challenge is how to judge the relative quality and quantity of work that individuals provide and to do so in a way that efficiently allocates rewards to millions of users. This requires the introduction of a scalable voting process. In particular it requires that authority to allocate funds must be as distributed and decentralized as possible. The first step in rewarding millions of users is to commit to distributing a fixed amount of currency regardless of how much work is actually done or how users vote. This changes the question from being “Should we pay?” to “Whom should we pay?” and signals to the market that money is being distributed and is being auctioned off to whoever “bids” the most work. This is similar to Bitcoin committing to award 50 BTC to whoever finds the most difficult hashes. Like Bitcoin, all work must be done prior-to payout and nothing should be paid speculatively on the promise to do work in the future. The next step is to reward everyone who does anything even remotely positive with something. This is accomplished by ranking all work done and distributing proportionally to its value. The more competitive the market becomes, the more difficult (higher quality or quantity) it becomes to earn the same payout.

Proof Of Hoof (POH) Overview:

At POH, we use Wrapped NFTs to tokenize real cow ownership on a decentralized blockchain, using DPoS technology. Users stake tokens, gaining influence in voting and resource allocation. Our rating system, 'Stasis,' cuts through the complexity, offering a straightforward measure of user contributions and impacts on the platform.

NFTs, acting as nodes, bring liquidity, with their reputation linked to 'Stasis.' We prioritize user control with flexible visibility options. The tokenomics mirrors the platform's interactive dynamics, shaping a dynamic market where user contributions directly affect their 'Stasis' values. We've gamified the process, making it engaging for our community. Please note, for in-depth mechanics, refer to the detailed technical documentation beyond the scope of this overview.

Blockchain Proofs:

History:

In the Stasis Rating System, history is not just a record; it's a dynamic force that ensures sustained benefits. Long-term staking of a specific cow becomes a testament to past contributions, providing participants with ongoing rewards and recognition. This historical record creates a meaningful connection between participants and their legacy within the ecosystem, fostering a sense of continuity and gratitude.

Stake:

Staking in our ecosystem goes beyond mere asset locking; it's a reputation-shaping mechanism. Particularly in the context of charitable contributions, stakes contribute to a positive public image and enhance participants' controlling power within the ecosystem. This reputation-driven approach not only benefits individuals but also leads to reduced gas consumption in trading activities, promoting a more sustainable and efficient network.

Work:

Innovative and eco-friendly, the concept of "work" within the POH ecosystem extends beyond conventional definitions. We harness the traction power of bulls for mining activities, showcasing a commitment to environmental consciousness. Moreover, our unique proof-of-work mechanism integrates agricultural processes with blockchain technology. By utilizing the anabolic digestion power from cow dung to generate gas for mining, we redefine sustainable mining practices, setting a precedent for green solutions in the blockchain space.

Addition layer of “intellectual rating”:

The system also includes a decentralized database of texts, such as laws, code, philosophy, and other important documents. Users will be able to mark specific parts of these texts and provide ratings based on their perceived importance, accuracy, relevance, or value. To ensure the integrity and reliability of the ratings, a weighted rating mechanism will be implemented. Ratings given by users with a high static reputation—based on their historical contributions and long-standing trust within the community—will carry more weight than those given by newer or less reputable users. This approach prevents any single user, especially those with limited reputation, from disproportionately influencing the perceived importance of texts. It also ensures that the evaluation of texts reflects the collective wisdom and trust of the community, promoting a fair and balanced representation of valuable content.

Ethics

To shortly illustrate what criteria may ethical rating be based on in connection with taking care of cow let us share something from book of Kenneth R. Valpey - ‘Cow Care in Hindu Animal Ethics’. Main thing is to approach cow’s not just mere NFTs objects, but real subjects with feelings and personality:

“This involves a change in the conception of citizenship that “recognizes that we are all interdependent, and experience varying forms and degrees of agency according to context, and over the life-course”. It then becomes possible to recognize that domestic animals (including bovines) possess, in their own ways, the three capacities specified as required for citizenship. Domestic animals demonstrate (1) the capacity to have a subjective good and the ability to communicate it; (2) the capacity to participate (specifically, by sheer presence in human settlement, as opposed to being made invisible, as in the modern meat industry); and (3) the capacity for cooperation, self-regulation, and reciprocity. Among nine specific areas of presupposition for citizenship discussed by the authors, we can give attention to four, with respect to cow care: (1) mobility and the sharing of public space; (2) use of animal products; (3) use of animal labor; and (4) sex and reproduction. “

- Kenneth R. Valpey - Cow Care in Hindu Animal Ethics

Economy

[Goldfinch Market Tail](#) (click for video):

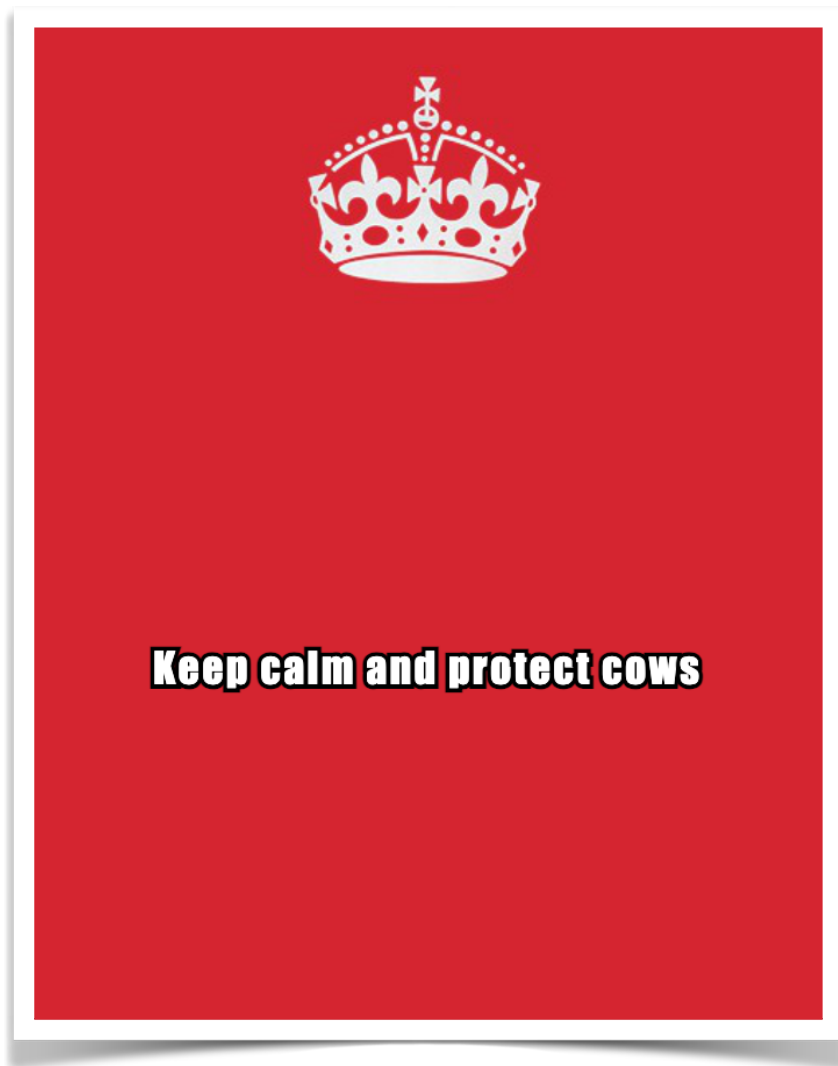
This video discusses the long-tail market strategy in cryptocurrency, highlighting the global accessibility of staking. This aligns with POH's approach, enabling even small stakeholders worldwide, support cow owners, and participate in a project.

[Ghee is From God:](#)

This video explores the divine and cultural aspects of ghee, emphasizing its market value. This understanding provides context to POH's cultural and ethical foundations, showcasing the economic significance of cow-related products.

Gas

"The Gobar Gas Research Station in Uttar Pradesh has established that one cow gives enough cow dung in a year to produce methane gas equivalent to 225 liters of petrol in energy terms. The calorific value table would show that one kg of methane gas is more or less equal in energy content to one kg of petrol, LPG, kerosene, or diesel." "LPG is generally used for cooking in urban areas while kerosene is the preferred fuel in rural India. A 15 kg LPG cylinder lasts about two months for a family of six. This works out to 15 kg of LPG per capita per year. The same holds true for kerosene. The entire LPG and kerosene requirements of our 100 crore population can be met by methane gas cylinders, produced from the cow dung of 75 million cows." "Just like CNG, methane gas can be used to run automobile engines in place of petrol. Our petrol consumption last year (2003-04) was eight million tonnes. On the assumption that one cow produces methane gas equivalent to 225 liters of petrol, we will need about 40 million cows to produce an energy equivalent to eight million tonnes of petrol. A generator needs 200 gm of petrol to produce one kilowatt/hour (kWh) of electrical energy. The per capita electrical energy consumption in rural areas is 112 kWh per annum. Our rural population being 74 crore, we will need another 85 million cows to meet the electrical energy needs of rural areas. This comes to a total of 200 million livestock to satisfy our energy requirements." (Dasa, Dr. Sahadeva. *Capitalism Communism And Cowism – A New Economics For The 21st Century*. Soul Science University Press. Kindle Edition).



Short AI Video summary